

1. Introduction

This report develops a relational data visualisation of product co-purchase patterns using the Online Retail II UCI dataset downloaded from Kaggle. The dataset contains transaction records from a UK-based non-store online retailer between December 2009 and December 2011. The original data is stored as a transactional table, where each row records a product purchased within an invoice. To make the data suitable for network visualisation, the transactions were cleaned and transformed in Python into product-product relationships. In the resulting graph, products are represented as nodes, while two products purchased in the same invoice are represented as weighted, undirected edges. Edge weight indicates how often each product pair was purchased together. The network was then visualised in Gephi to explore product hubs, co-purchase communities, strong relationships, peripheral products, and scalability issues. The aim is to evaluate how effectively a network-based visualisation can reveal relational structure and support exploratory visual analysis of transaction data.

2. Dataset Preparation and Graph Modelling

2.1 Dataset and cleaning

The dataset used in this project was the Online Retail II UCI dataset, downloaded from Kaggle. It contains product-level transaction records from a UK-based online retail business. The raw CSV file included fields such as Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID, and Country (Miyabon, 2019). Since the original data was stored as a transactional table, it required cleaning and transformation before it could be used for relational network visualisation.

Python was used to prepare the dataset. Cancelled invoices were removed because they did not represent completed purchases. Records with invalid quantities or prices were

also excluded to avoid distorted co-purchase relationships. Rows with missing invoice, product code, or product description fields were removed because these fields were required to identify valid products and transactions. After cleaning, products were grouped by invoice to form customer baskets for graph construction.

2.2 Graph modelling

After cleaning, the transaction table was transformed into a relational graph. Each invoice was treated as a customer basket containing the products purchased together. For every basket with at least two products, Python generated all possible product-product pairs. If two products appeared in the same invoice, an edge was created between them. When the same pair appeared in multiple invoices, the edge weight increased. This allowed repeated co-purchase relationships to be represented quantitatively rather than only as simple links.

In this graph model, each product or StockCode was represented as a node. An edge represented a co-purchase relationship between two products that appeared in the same invoice. The edge weight represented the number of repeated co-purchases between the two products across invoices. The final graph was therefore a weighted, undirected product co-purchase network. It was undirected because the relationship between two products is mutual: if product A is purchased with product B, then product B is also purchased with product A. The processed node and edge lists were then exported as CSV files and imported into Gephi for visualisation.

3. Visualisation Method and Implementation

3.1 Python Workflow

Python was used to transform the cleaned transaction data into a format suitable for network visualisation. After products were grouped by invoice, the code generated product-product pairs from each valid basket. Repeated pairs were counted to create a weighted edge list, where weight represented the number of invoices in which two products appeared together. A node list was also created for the products included in the filtered network. To reduce visual clutter, edge-weight thresholds were applied. The main visualisation used the threshold-100 network, which retained product pairs purchased together at least 100 times.

3.2 Gephi Workflow

The processed node and edge CSV files were imported into Gephi as a weighted, undirected graph. The threshold-100 network contained 581 product nodes and 1,948 co-purchase edges. Gephi was selected because it supports interactive network exploration, graph layout algorithms, community detection, and visual encoding of node and edge attributes (Bastian et al., 2009). The graph was arranged using ForceAtlas2, a force-directed layout algorithm for network visualisation (Jacomy et al., 2014). In this project, the layout was configured with scaling set to 2.0, gravity set to 1.0, and edge weight influence set to 1.0. Noverlap was then applied separately to reduce node overlap after the main layout process.

3.3 Visual Encodings

The visualisation used several encodings to represent network structure. Each node represented a product, and each edge represented a co-purchase relationship between two

products. Node size was based on degree, so larger nodes indicated products connected to more co-purchased products. Modularity-based community detection was used to identify product communities in the network (Blondel et al., 2008). Node colour was then based on modularity class, so products in the same detected community shared the same colour. Edge colour intensity was based on weight, so stronger co-purchase relationships appeared more visually prominent. Modularity analysis produced a modularity score of 0.679 and 79 detected communities.

3.4 Interaction

The Gephi workspace supported interactive exploration before the figures were exported. Zooming and panning were used to inspect dense graph areas that were not readable in the full network overview. Node selection and the Data Laboratory were used to inspect product attributes, identify high-degree products, and extract the hub products reported in Table 1. Edge-weight filtering was also used interactively to compare the threshold-100 and threshold-30 networks.

These interactive steps supported the analysis rather than only the presentation of the graph. The threshold-100 network was used for focused interpretation because it retained strong repeated co-purchase relationships, while the threshold-30 network was used to demonstrate how lower filtering levels increase density and visual clutter. Therefore, interaction helped identify product hubs, inspect strong co-purchase relationships, and evaluate the scalability of the node-link method.

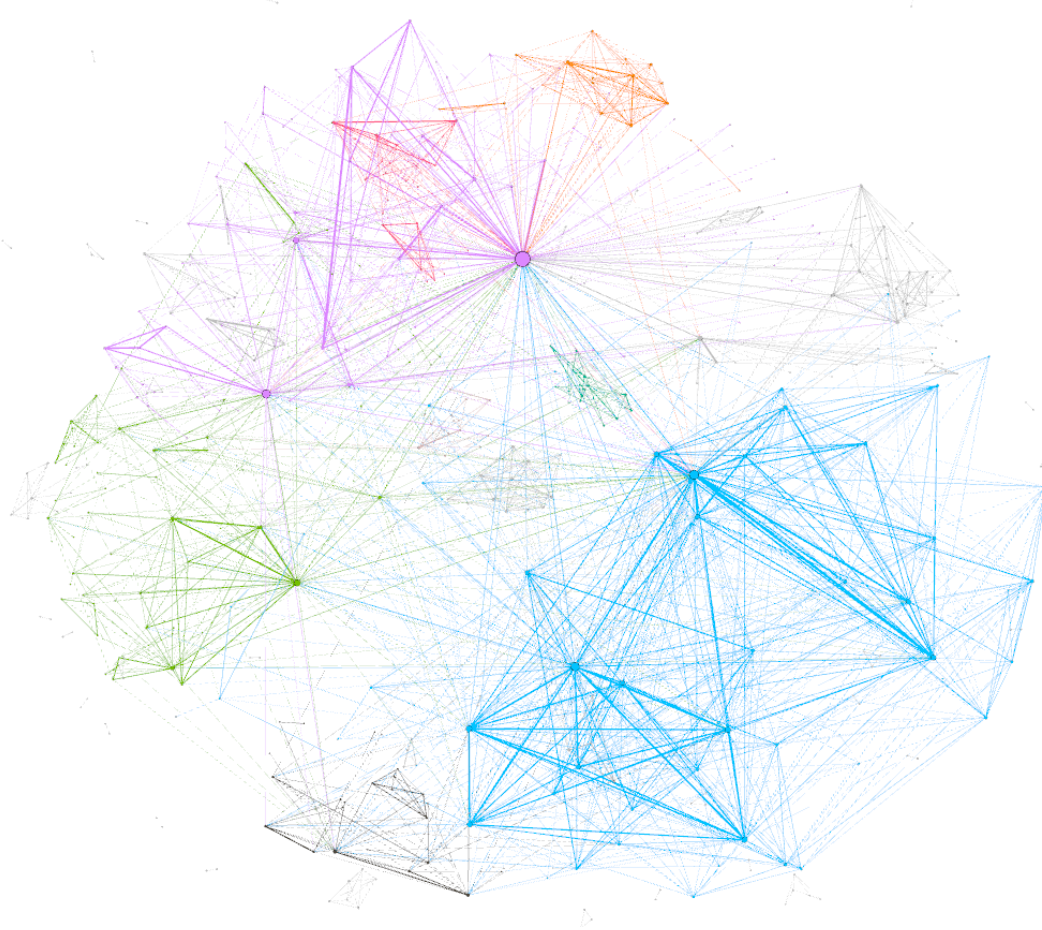
4. Visualisation Results

4.1 Threshold-100 Network Overview

Figure 1 presents the main threshold-100 product co-purchase network. This view retains product pairs purchased together at least 100 times, reducing weaker relationships and improving readability. The network shows several visible communities and a small number of larger nodes, suggesting both clustered product groups and hub products.

Figure 1

Threshold-100 Product Co-Purchase Network Overview



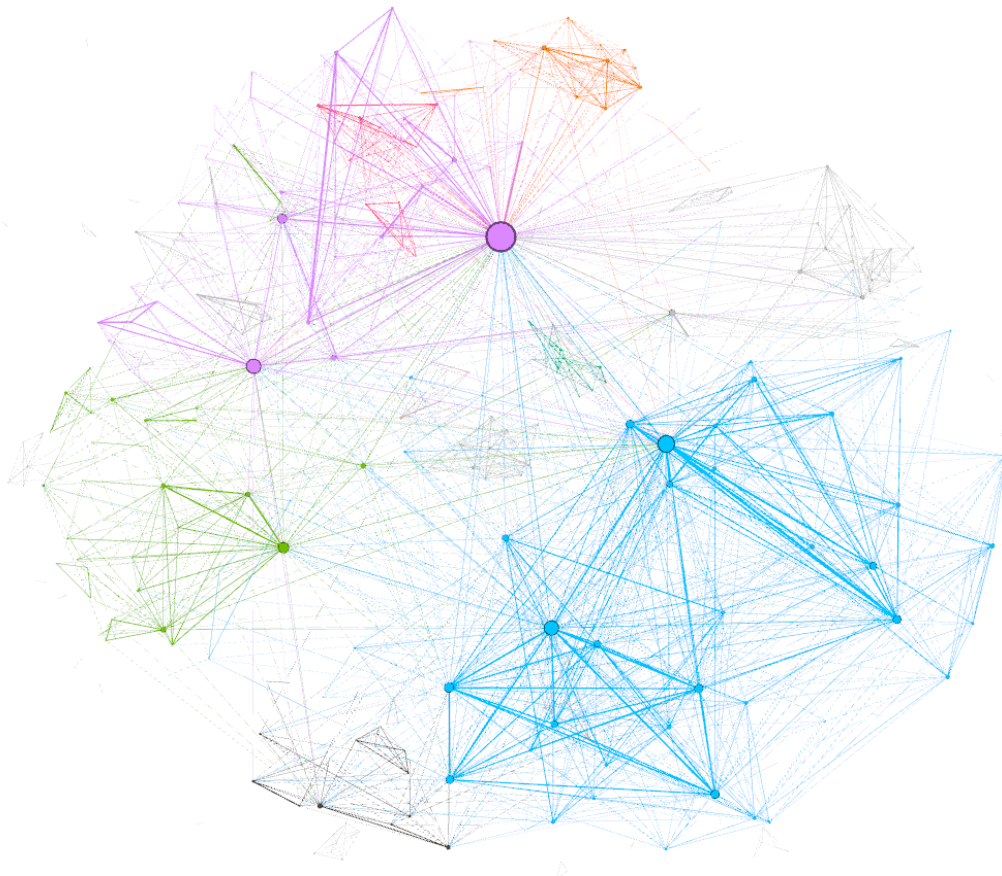
Note. Nodes represent products and edges represent co-purchase relationships. Node size represents degree, node colour represents modularity class, and edge colour intensity reflects edge weight. The threshold-100 network contains 581 product nodes and 1,948 weighted co-purchase edges. The graph was arranged in Gephi using ForceAtlas2 and Noverlap.

4.2 Product Hubs

Figure 2 highlights product hubs by increasing the visual emphasis on node degree. Larger nodes represent products connected to more co-purchased products. This view makes the hub structure more visible and supports later analysis of high-degree products.

Figure 2

Product Hubs Based on Degree in the Co-Purchase Network



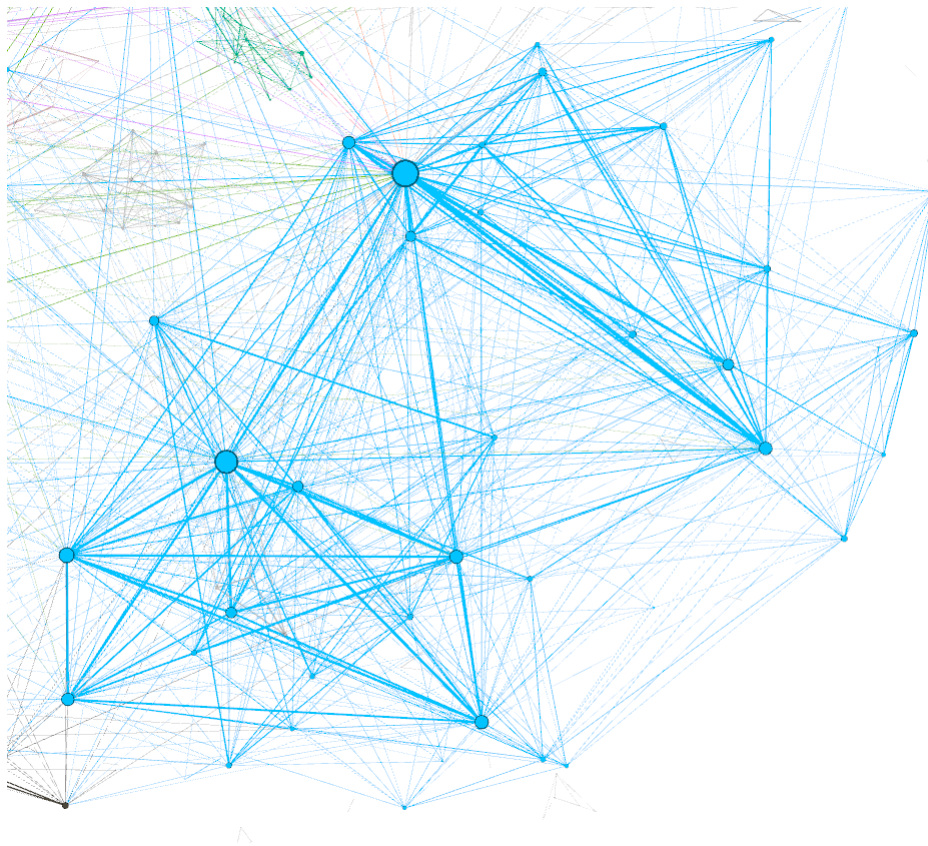
Note. Node size represents degree. Larger nodes indicate products connected to more co-purchased products and can be interpreted as product hubs. Node colour represents modularity class, while edges represent co-purchase relationships in the threshold-100 network.

4.3 Strong Co-Purchase Relationships

Figure 3 zooms into a dense community within the threshold-100 network. This focused view makes strong product-pair relationships easier to inspect because more visible edges indicate higher repeated co-purchase frequency.

Figure 3

Strong Co-Purchase Relationships Within a Product Community



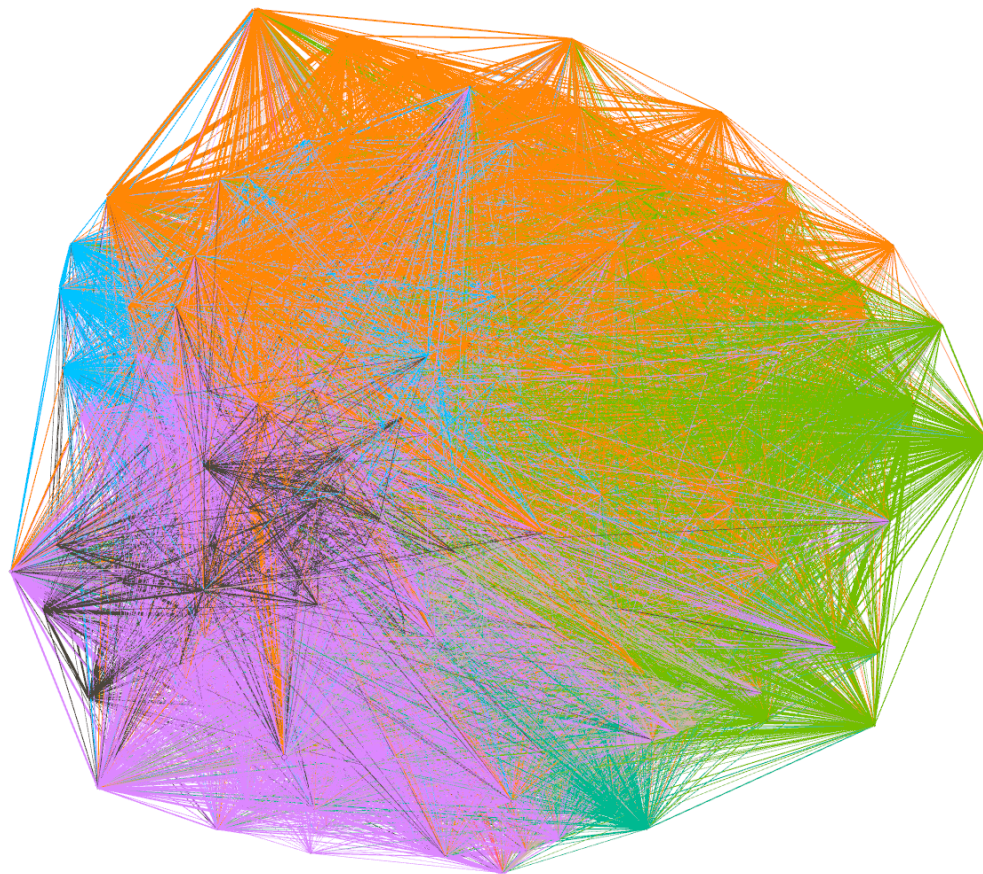
Note. The figure zooms into a dense area of the threshold-100 network. Node colour represents modularity class, and edge colour intensity reflects edge weight. More visible edges indicate product pairs that were purchased together more frequently.

4.4 Threshold Comparison and Visual Clutter

Figure 4 shows the threshold-30 network. Compared with the threshold-100 network, it contains more products and edges, creating a dense hairball effect. This demonstrates that lower edge-weight thresholds increase coverage but reduce interpretability.

Figure 4

Visual Clutter in the Threshold-30 Co-Purchase Network



Note. The threshold-30 network contains 1,653 product nodes and 20,065 co-purchase edges, making it substantially denser than the threshold-100 network. Although node colours indicate modularity-based communities, the large number of overlapping edges creates a hairball effect and reduces interpretability.

5. Analysis Results and Findings

5.1 Overall Network Structure

The visualisations show that the product co-purchase network does not appear evenly distributed. In Figure 1, products are grouped into several visible communities, with some groups more densely connected than others. This suggests that repeated co-purchases are organised around particular product groups or basket patterns. The network also shows a core-periphery-like structure, where several dense communities are connected to the wider network, while smaller groups remain more peripheral. This indicates that not all products play the same role in the co-purchase structure.

5.2 Product Hubs

Figure 2 highlights the presence of product hubs based on degree. Larger nodes represent products connected to more co-purchased products, meaning these products appear with a wider range of other items in customer baskets. These hub products may act as anchors within the network because they connect multiple products or communities. From an exploratory retail perspective, such products may provide starting points for cross-selling analysis. However, this interpretation should remain cautious because degree only shows network connectivity, not profitability or causal influence.

To support the visual interpretation in Figure 2, the highest-degree products in the threshold-100 network were extracted from the node list.

Table 1*Top Product Hubs in the Threshold-100 Network*

Rank	StockCode	Product Description	Degree
1	85123A	WHITE HANGING HEART T-LIGHT HOLDER	155
2	85099B	JUMBO BAG RED WHITE SPOTTY	89
3	20725	LUNCH BAG RED SPOTTY	77
4	22423	REGENCY CAKESTAND 3 TIER	77
5	21212	PACK OF 72 RETRO SPOT CAKE CASES	56

Note. Degree represents the number of unique products connected to each product in the threshold-100 co-purchase network. Higher degree values indicate products that were co-purchased with a wider range of other products.

5.3 Co-Purchase Communities

The modularity analysis detected 79 communities with a modularity score of 0.679, which supports the visual pattern shown in Figure 1. The presence of multiple communities suggests that products tend to form repeated co-purchase groups. These communities may reflect product families, product themes, or common purchase missions. However, the community size distribution is uneven: a small number of communities are visually dominant, while many others are smaller and more peripheral. Therefore, the findings should focus on the major visible communities rather than attempting to interpret all detected communities individually.

5.4 Strong Relationships and Irregularities

Figure 3 shows that strong co-purchase relationships are concentrated within dense product communities. To support this visual interpretation, the highest-weight product pairs were extracted from the edge list.

Table 2

Top Strong Co-Purchase Pairs in the Product Network

Rank	Source StockCode	Target StockCode	Edge Weight
1	22386	85099B	914
2	21733	85123A	805
3	21931	85099B	761
4	82482	82494L	759
5	85099B	85099F	725

Note. Edge weight represents the number of invoices in which the two products were purchased together.

Table 2 shows that the strongest product-pair relationships appeared together in hundreds of invoices. These pairs may be useful exploratory signals for possible bundles or recommendations, although co-purchase frequency shows association rather than causation. Figure 4 reveals a key limitation of the method. When the edge-weight threshold is reduced to 30, the network becomes much denser and produces a hairball effect.

Table 3*Threshold Comparison of the Product Co-Purchase Networks*

Network	Threshold	Nodes	Edges	Main finding
Threshold-100 network	≥ 100	581	1,948	Cleaner and more readable
Threshold-30 network	≥ 30	1,653	20,065	Dense hairball effect

Note. Lowering the edge-weight threshold increases the number of nodes and edges substantially. This improves coverage of weaker co-purchase relationships but reduces readability because of edge overlap and visual clutter.

Table 3 shows that threshold selection directly affects interpretability. The threshold-100 network is more suitable for focused analysis, while the threshold-30 network is useful for demonstrating scalability limits. Overall, the findings show that network visualisation can identify hubs, communities, and strong relationships, but the results require filtering and critical interpretation.

6. Method Evaluation and Scalability

6.1 Advantages of the Chosen Method

The node-link network diagram was suitable for this project because it directly represented product-product relationships. Products could be shown as nodes, while co-purchase relationships could be shown as weighted edges. This made hubs, communities, dense areas, and peripheral groups easier to identify than in the original transaction table. Gephi also supported interactive exploration, including zooming, panning, node selection, attribute inspection, and filtered network comparison (Bastian et al., 2009). ForceAtlas2 was

appropriate because the dataset did not have a fixed hierarchy; the layout allowed connected products to move closer together while separating less connected products (Jacomy et al., 2014).

6.2 Disadvantages and Limitations

Despite these advantages, node-link diagrams have clear limitations. As the number of nodes and edges increases, the visualisation can become cluttered because many edges overlap (Ghoniem et al., 2005). This was visible in the threshold-30 network, where the lower threshold included many more co-purchase relationships and created a hairball effect. Another limitation is that the layout is algorithm-dependent because ForceAtlas2 computes node positions through a layout algorithm rather than from original business variables (Jacomy et al., 2014). The position of a node in ForceAtlas2 is not an original data value, so spatial distance should be interpreted carefully. A product placed near the centre of the layout may be structurally important, but its position does not directly represent revenue, profit, or business value. In addition, co-purchase relationships show association only. They do not prove that one product caused another product to be purchased.

6.3 Comparison With Other Relational Visualisation Methods

Compared with other relational visualisation methods, the node-link network was the most suitable primary method for this project. An adjacency matrix could represent product-product relationships and may work better for dense graphs because it avoids edge crossing, although it is less intuitive for visually identifying hubs and communities (Ghoniem et al., 2005). Tree layouts were less suitable because the product co-purchase network is not hierarchical; products can have many-to-many relationships and cycles. Treemaps or enclosure methods could show product or category size effectively, but they are designed for

hierarchical structures rather than direct product-product relationships (Shneiderman, 1992). Sankey diagrams are useful for representing flow relationships, but this dataset represents unordered product baskets rather than sequential flows or customer journeys (Riehmann et al., 2005).

6.4 Scalability for Large Relational Datasets

The method is partially scalable. It works well for filtered small to medium networks, but it becomes less effective for dense relational datasets because edge overlap reduces readability (Ghoniem et al., 2005). The threshold comparison showed this clearly: the threshold-100 network remained interpretable, while the threshold-30 network became visually dense. Therefore, node-link diagrams require filtering, aggregation, community detection, and interaction when applied to larger relational datasets.

7. Critical Reflection and Conclusion

Although the visualisation revealed meaningful exploratory co-purchase patterns, the results should be interpreted critically. Co-purchase does not imply causality because products may appear together due to promotions, seasonality, price changes, stock availability, or ordering behaviour. The analysis was also limited by missing business context, such as customer segments, product margins, sales channels, and time-based changes.

Overall, this project transformed Online Retail II transaction data into a weighted, undirected product co-purchase network using Python and Gephi. The visualisation helped identify hubs, communities, strong relationships, and scalability issues. The method is useful for exploratory relational visual analytics, but it requires filtering, interaction, and business validation to remain interpretable.

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